Exact K2P and K3P Tree–Theta-Trinet Collisions

Technical summary

Alec Kriebel ORCID: 0009-0001-9320-500X August 2026

Exact finding. A binary semi-directed strict level-two theta trinet with a genuine nontrivial 3-blob shares an exact K2P distribution with a three-star tree. Under the conventions stated in arXiv:2607.12919v2, this appears to conflict with the conclusion of Lemma 5.6 and with the K2P case of Corollary 5.8. The present construction does not test the paper's JC or level-one results. A companion exact K3P collision answers its open high-level K3P trinet question negatively.

Common topology. Use

$$\rho \to 1, \rho \to u,\ u \to p, q,\ p, q \to r_2,\ p, q \to r_3,\ r_2 \to 2,\ r_3 \to 3,$$

with both inheritance probabilities 1/2. Root suppression gives paths $p - u - q$, $p - r_2 - q$, $p - r_3 - q$ and three incident leaf edges. For the explicit symmetric witnesses, U labels $u \to p$, V labels $u \to q$, S labels both $p \to r_2, p \to r_3$, and T labels both $q \to r_2, q \to r_3$; the other four arcs carry K. The rank and implicit-function calculations instead use independent parameters on all nine effective semi-directed edges and both inheritance probabilities.

Simple exact K2P witness. In order (A, C, G, T), assign

$$K = (1, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), \quad U = (1, \tfrac{4}{5}, \tfrac{19}{30}, \tfrac{4}{5}), \quad V = (1, \tfrac{7}{240}, \tfrac{239}{360}, \tfrac{7}{240}),$$
$$S = (1, \tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}), \quad T = (1, \tfrac{1}{3}, \tfrac{1}{27}, \tfrac{1}{3}).$$

Use K on the four common K-edges $\rho \to 1, \rho \to u, r_2 \to 2, r_3 \to 3$, with the explicit U, V, S, T placement above. Every transition entry is positive; the smallest is 1/120. For $x = y + z$, these common edges contribute $K_x K_{y+z} K_y K_z = K_x^2 K_y K_z$. The four retained-parent choices $(p, p), (p, q), (q, p), (q, q)$ give, respectively, the four terms in

$$M_{y,z} = \tfrac{1}{4}(S_y S_z U_x + S_y T_z U_y V_z + T_y S_z U_z V_y + T_y T_z V_x).$$

With $s = \sqrt{71}$,

$$P = (1, \tfrac{151}{36s}, \tfrac{107}{162}, \tfrac{151}{36s}), \quad R = (1, \tfrac{s}{40}, \tfrac{31}{120}, \tfrac{s}{40}),$$

and exact arithmetic gives $M_{y,z} = P_{y+z} R_y R_z$. The tree vectors are

$$\alpha = (1, \tfrac{151s}{10224}, \tfrac{107}{648}, \tfrac{151s}{10224}), \quad \beta = \gamma = (1, \tfrac{s}{80}, \tfrac{31}{240}, \tfrac{s}{80}).$$

All 64 Fourier and pattern probabilities agree exactly; the minimum is $1188799/79626240 = 0.01492973924\ldots$. The source invariant Q is zero. A graph-based verifier literally deletes the two unselected reticulation arcs, reconstructs the four monomials, and independently confirms all 64 probabilities by ordinary-state Markov pruning.

Proof diagnosis. The level-one sign factorization is valid after placing the active reticulation leaf in a designated position. In the level-two induction a child can retain a reticulation adjacent to another leaf. Positivity after a child-specific relabelling does not imply positivity of the fixed-order child invariant in the parent expansion. On the exact continuous-time witness, one child has $Q = -1.919971072382827\ldots \times 10^{-9}$ in the parent order and $Q = 3.428488326525925\ldots \times 10^{-9}$ after the favorable swap. A separate fully specified rational theta point has $Q < 0$ in all six orders.

K2P geometry and continuous time. The affine K2P space has dimension 9 and the tree model dimension 6. At the simple collision, the fixed theta map has determinant

$$-\frac{7^2 11^2 19\, 107\, 151^2\, 15013}{2^{60} 3^{25} 5^{10}} \neq 0.$$

It also has rank 9 at the strict continuous-time collision. Hence the local semi-directed theta collision locus has dimension 17 and codimension 3: generic theta parameters are distinguishable, but full distinguishability fails. An exact symmetric ansatz contains a local six-dimensional collision family. The strict continuous-time witness satisfies $g > s^2$ on every edge, has minimum eigenvalue 1/16 and minimum rate margin 11/900, and is verified by direct Markov pruning.

K3P companion. Let $h = 5^{-1/4}$. Use the same fixed edge placement and the same four-switching matrix M, now with

$$U = (1, h/3, h, 1/3),\ V = (1, h, h/3, 1/3),\ S = (1, 3h^2/4, 1/4, 3/10),\ T = (1, 1/4, 3h^2/4, 3/10).$$

This mixture factors with

$$B = (1, h^2/2, h^2/2, h^2/2), \quad P = (1, (5h^3 + h)/4, (5h^3 + h)/4, h^2).$$

The fixed theta map has rank 15 with

$$\det J_* = \frac{h(10h^2 + 1)}{2^{61} 3^4 5^{14}} > 0.$$

Its local collision locus has dimension 23 and codimension 6. A real-analytic implicit-function argument, with Jacobian and tangent data verified exactly, moves the collision into the cone where all three K3P rate classes are positive.

Status and replay. The canonical manuscript, exact certificates, and verifier package are available in the versioned GitHub snapshot (v1.0.0). With Python 3.10 or newer, run `python3 verify_k2p_displayed_trees.py` for the graph reconstruction and direct-pruning audit, or `python3 verify.py` for every check; the latter ends with `ALL EXACT CHECKS PASSED`.